

SYNTHEA HEALTHCARE DATA ANALYSIS

Phase 2: Patient Demographics & Healthcare Expenses

OVERVIEW

1. Project Background and Purpose

This project analyzes the synthetic electronic health records (EHR) data that's generated by Synthea to explore the patients' demographics, healthcare utilization, health conditions, and healthcare expenses using SQL.

2. Dataset Overview

Data Source	Synthea
Database	MySQL
Current Table	Patients
Patient Count	116
Data Type	Synthetic EHR data

PATIENT DEMOGRAPHICS

PATIENT POPULATION

1.1 Patient Population

Research question: How many patients are included in the dataset?

Findings: The dataset included 116 synthetic patients.

1.2 Gender Distribution

Research question: What is the gender distribution of the patient population?

Findings: Female patients represented the larger group of the dataset, with 64 patients compared with 52 male patients

Gender	Number of Patients
Female	64
Male	52
Total	116

1.3 Racial Distribution

Research Question: What is the racial distribution of the patient population?

Interpretation: The synthetic dataset was predominantly composed of patients categorized as White, with considerably smaller numbers of Asian, Black, and Native patients. Because the dataset is synthetic, these distributions should not be interpreted as representative of a real-world population.

Race	Number of Patients
White	104
Asian	7
Black	4
Native	1
Total	116

1.4 Patient Age

Research Question: What is the average age of patients in the dataset?

Method: Patient age was calculated from the BIRTHDATE field using December 31, 2025, as the reference date.

Finding: The average age of patients was 42.65 years.

1.5 Age Distribution

Research Question: How are patients distributed across age groups?

Result Grid			Filter Rows:
	age_group	patient_count	
▶	0-17	19	
	18-34	26	
	35-49	26	
	50-64	21	
	65+	24	

Interpretation: Patients were represented across all five age categories. The 18–34 and 35–49 groups were the largest, with 26 patients each, while the 0–17 group was the smallest, with 19 patients.

HEALTHCARE EXPENSES

COST ANALYSIS

2.1 Average Healthcare Expenses

Research Question: What are the average healthcare expenses among patients?

Finding: The average healthcare expense was \$714,361.46 per patient.

avg_healthcare_expenses	avg_healthcare_coverage	avg_uncovered_expenses
714361.46	7460.26	706901.2

2.2 Healthcare Expenses by Gender

Research Question: Do average healthcare expenses differ by gender?

Findings: Male patients had average healthcare expenses of \$711,634.27, while female patients had average healthcare expenses of \$716,573.30.

Interpretation: Average healthcare expenses differed between male and female patients in the synthetic dataset. However, the difference between the two groups was relatively small.

	GENDER	ROUND(AVG(HEALTHCARE_EXPENSES), 2)
▶	M	711634.27
	F	716577.3

2.3 Patients with the Highest Healthcare Expenses

Research Question: Which patients in the dataset had the highest healthcare expenses?

Method: Patients were ranked by total healthcare expenses in descending order, and the ten patients with the highest recorded expenses were identified.

	Id	FIRST	LAST	HEALTHCARE_EXPENSES
▶	799d0738-d5cd-4161-bdb8-e3c49e21ec66	Sammie902	Greenholt190	1798183.1700000002
	8d1ba4bb-7250-4295-be1c-5d0d423e55f7	Orlando897	Reichert620	1681686.84
	53d371cc-eb68-4740-a68c-39707c6cb519	Stevie682	Hilpert278	1608750.03
	8abe8efb-d431-4465-ad3b-7b3f301b845f	Maisha241	Bartell116	1566968.11
	1d7fe784-8667-47e8-b733-8e6ec0fbb5f9	Tia76	Crona259	1559999.46
	e4f1bd35-93ae-418b-b8cf-d39c3fbb069c	Raye931	Wunsch504	1551125.51
	8b17ff1e-2b24-46a9-83f7-f8a49e82c604	Emerson869	Hagenes547	1541995.6
	be249d5b-327e-4c30-8fb2-3a1c587fe5ad	Kermit56	Rutherford999	1537567.36
	c1b80a65-61e0-4f1b-9e1f-122c6d326034	Sabina296	Lind531	1481953.82
	9fe859fa-d775-494d-ae34-fa648e5d186d	Erika442	Becker968	1477931.49

Finding: The ten patients with the highest healthcare expenses ranged from approximately \$1.48 million to \$1.80 million. The highest recorded healthcare expense was approximately \$1.80 million.

Interpretation: The results demonstrate substantial healthcare expenses among the highest-cost patients in the synthetic population. Further analysis of patient encounters and health conditions may help identify factors associated with higher healthcare expenses.

2.4 Healthcare Expenses by Age Group

Research Question: How do average healthcare expenses differ across patient age groups?

Method: Patients were categorized into five age groups (0–17, 18–34, 35–49, 50–64, and 65+). The average healthcare expenses were then calculated for each age group.

	age_group	avg_healthcare_expenses
▶	0-17	94747.45
	18-34	407012.81
	35-49	759526.33
	50-64	953117.99
	65+	1280009.67

Finding: Average healthcare expenses varied considerably across age groups. Patients aged 65 and older had the highest average healthcare expenses at approximately \$1.28 million, followed by patients aged 50–64 at approximately \$953,118 and patients aged 35–49 at approximately \$759,526. The 0–17 age group had the lowest average healthcare expenses at approximately \$94,747.

Interpretation: The results show a general pattern of increasing healthcare expenses across older age groups in this synthetic patient population. This suggests that age may be associated with higher healthcare expenses within the dataset. Additional analysis of healthcare encounters and health conditions may help explain the differences observed between age groups.

2.5 Healthcare Expenses and Healthcare Coverage

Research Question: How does average healthcare coverage compare with average healthcare expenses?

Method: Average healthcare expenses and average healthcare coverage were calculated across the patient population. The difference between these two fields was also calculated to compare their values within the dataset.

avg_healthcare_expenses	avg_healthcare_coverage	avg_uncovered_expenses
714361.46	7460.26	706901.2

Finding: Patients had average healthcare expenses of approximately \$714,361.46, while average healthcare coverage was approximately \$7,460.26. The average difference between healthcare expenses and healthcare coverage was approximately \$706,901.20.

Interpretation: The analysis identified a substantial difference between the healthcare expense and healthcare coverage values recorded in the synthetic dataset. Healthcare expenses were considerably higher than healthcare coverage on average. Further investigation of the dataset's payer and encounter information would be necessary to better understand how healthcare costs and coverage are represented.

HEALTHCARE UTILIZATION ENCOUNTER ANALYSIS

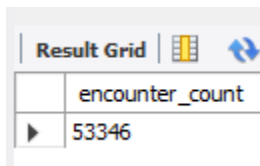
Phase 3: Healthcare Utilization & Encounter Analysis

ENCOUNTER OVERVIEW

3.1 Total Healthcare Encounters

Research Question: How many healthcare encounters are recorded in the dataset?

Findings: The dataset contains 53,346 healthcare encounters across the synthetic patient population.



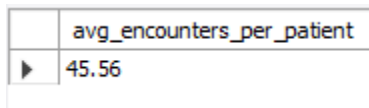
Result Grid	
	encounter_count
▶	53346

Interpretation: The encounter dataset contains more records than the patient dataset because individual patients can have multiple healthcare encounters over time. This provides the foundation for examining differences in healthcare utilization across patients and demographic groups.

3.2 Encounters per Patient

Research Question: What is the average number of encounters per patient?

Method: Healthcare encounters were grouped by patient ID to determine each patient's total number of encounters. The average was then calculated across patients represented in the encounter data.



	avg_encounters_per_patient
▶	45.56

Findings: Patients represented in the encounter data had an average of 45.56 healthcare encounters per patient.

Interpretation: The results indicate that patients had multiple healthcare encounters recorded over time. However, encounter frequency varied substantially across individuals, suggesting differences in healthcare utilization within the synthetic patient population.

3.3 Most Common Encounter Types

Research Question: What are the most common types of healthcare encounters in the dataset?

Findings: Wellness encounters were the most common, with 19,106 encounters, followed by ambulatory encounters with 18,936. Outpatient encounters were the third most common at 9,003. Urgent care, emergency, and inpatient encounters occurred less frequently.

	ENCOUNTERCLASS	encounter_count
►	wellness	19106
	ambulatory	18936
	outpatient	9003
	urgentcare	2373
	emergency	2090
	inpatient	1838

Interpretation: The encounter distribution shows that a large portion of recorded healthcare utilization consisted of wellness and ambulatory care. More intensive encounter types, such as emergency and inpatient care, represented a smaller portion of the encounters in this synthetic population.

3.4 Patients with the Highest Healthcare Utilization

Research Question: Which patients have the greatest number of healthcare encounters?

Findings: The patient with the highest recorded healthcare utilization had 2,006 encounters, followed by patients with 1,606 and 1,538 encounters. The remaining patients in the top had between 346 and 826 encounters.

	PATIENT	encounter_count
►	cae10920-f977-48b4-a0d3-4d70ad561fd1	2006
	19d2cfb8-439b-454a-b47e-5274c219005b	1606
	3f336702-bf73-4fc8-bd59-3ba77fd65d0d	1538
	3acf9313-1874-4dff-ab2a-3187516d92d6	826
	cecb7ece-fb70-4a7f-b51d-6e8dd1506ad9	452
	3dd2dd29-7cd0-48f7-b859-136c6e9d36e0	385
	714b9c18-783d-4f52-aa64-cc3a05a286d9	375
	6ec18ddf-e9ee-421a-9033-456f558c7b4b	367
	2c71dd97-7085-416a-aa07-d675bbe3adf2	367
	137acc1b-dbca-473e-84bb-fe1baaf98819	346

Interpretation: Healthcare utilization varied substantially among patients. A small number of patients accumulated particularly high numbers of encounters, suggesting that utilization was not evenly distributed across the synthetic patient population. Further analysis of encounter types and health conditions may help explain these high encounter frequencies.

3.5 Encounter Costs

Research Question: What is the average cost of a healthcare encounter in the dataset?

Method: Average base encounter cost and average total claim cost were calculated across all healthcare encounters. The two cost fields were also compared at the record level to determine whether any encounters contained different values.

	different_cost_records
▶	0

Findings: The average base encounter cost and average total claim cost were both \$128.75. No records contained different values between the two cost fields.

	avg_base_encounter_cost	avg_total_claim_cost
▶	128.75	128.75

Interpretation: The base encounter cost and total claim cost were identical across all recorded encounters. Therefore, these variables do not provide distinct measures of encounter cost in this dataset. The average recorded encounter cost was \$128.75.

3.6 Healthcare Utilization by Age Group.

Research Question: How does healthcare utilization vary across patient age groups?

Method: Patient demographic data were joined with healthcare encounter records using the unique patient identifier. Patient age was calculated using a fixed reference date of December 31, 2025, and patients were categorized into the same age groups used in the demographic analysis. The average number of encounters per patient was then calculated for each age group.

Findings: Patients aged 65 and older had the highest average healthcare utilization, with 49.75 encounters per patient. Patients aged 50-64 averaged 37.52 encounters, followed by ages 18-34 at 35.85 and ages 35-49 at 34.62. Patients aged 0-17 had the lowest average utilization at 16.32 encounters per patient.

	age_group	avg_encounters
▶	0-17	16.32
	18-34	35.85
	35-49	34.62
	50-64	37.52
	65+	49.75

Interpretation: Healthcare utilization generally differed by age group within the synthetic population. The oldest age group demonstrated the highest average utilization, while the youngest group demonstrated substantially lower utilization. However, utilization did not increase consistently across every successive age category, as the three middle age groups had relatively similar averages. Further analysis of patients' documented health conditions may help identify factors associated with these utilization differences.

HEALTH CONDITIONS CLINICAL ANALYSIS

Phase 4: Health Conditions and Clinical Analysis

CONDITION OVERVIEW

4.1 Most Common Health Conditions

Research Question: What are the most documented health conditions in the patient population?

Method: Condition records were grouped by their clinical description and counted to identify the ten most frequently documented conditions in the dataset.

Findings: Viral sinusitis was the most frequently documented condition, with 1,248 records. This was followed by acute viral pharyngitis with 653 records and acute bronchitis with 563 records. Other frequently documented conditions included normal pregnancy, obesity, prediabetes, hypertension, anemia, chronic sinusitis, and miscarriage in the first trimester.

	DESCRIPTION	condition_count
▶	Viral sinusitis (disorder)	1248
	Acute viral pharyngitis (disorder)	653
	Acute bronchitis (disorder)	563
	Normal pregnancy	516
	Body mass index 30+ - obesity (finding)	449
	Prediabetes	317
	Hypertension	302
	Anemia (disorder)	300
	Chronic sinusitis (disorder)	236
	Miscarriage in first trimester	221

Interpretation: The results show that respiratory conditions were among the most frequently documented conditions in the dataset, particularly viral sinusitis, acute viral pharyngitis, and acute bronchitis. The Top 10 also included chronic or metabolic conditions such as obesity, prediabetes, and hypertension, indicating that the condition records represent a mixture of acute and longer-term health concerns.

4.2 Conditions per Patient

Research Question: What is the average number of documented health conditions that are recorded per patient?

Method: Condition records were grouped by patient ID to determine the number of documented conditions recorded per patient. The average was then calculated across patients represented in the conditions dataset.

Findings: Patients represented in the conditions data had an average of 7.27 documented condition records per patient.

	avg_conditions_per_patient
▶	7.27

Interpretation: The results indicate that patients commonly had multiple condition records documented in the synthetic EHR data. However, this measure reflects the number of condition records, rather than the number of unique diagnoses, because the same condition may be documented more than once for an individual patient.

4.3 Patients with Multiple Conditions

Research Question: How many patients have multiple distinct documented health conditions?

Method: Condition records were grouped by patient ID, and distinct condition descriptions were counted for each patient. Patients with more than one distinct documented condition were classified as having multiple conditions. The number of these patients was compared with the total number of unique patients represented in the condition's dataset.

Finding: Of the 1,152 unique patients represented in the conditions dataset, 1,115 had more than one distinct documented health condition, representing approximately 96.8% of patients in the table.

	total_patients_with_conditions		patients_with_multiple_conditions
▶	1152	▶	1115

Interpretation: Most patients represented in the conditions dataset had multiple distinct health conditions documented. This indicates a high prevalence of multiple-condition records within this synthetic sample. However, the result should be interpreted specifically within the conditions dataset and should not be generalized to the separate 116-patient population analyzed earlier.

4.4 Conditions and Healthcare Utilization

Research Question: How is the number of distinct health conditions associated with healthcare utilization?

Method: The number of documented conditions was calculated for each patient and combined with each patient's total number of healthcare encounters. Patients were then grouped according to their number of unique conditions, and the average number of encounters was calculated for each group. Patient-level totals were calculated separately before joining the datasets to prevent duplicate records from inflating conditions or encounter counts.

Findings: Average healthcare utilization generally increased among patients with a higher number of distinct documented conditions. Patients with one unique condition averaged 11.27 encounters, while those with five conditions averaged 30.78, and those with ten conditions averaged 63.15. Higher condition-count groups showed greater variability and generally contained fewer patients.

	unique_conditions	patient_count	avg_encounters
▶ 1	1	37	11.27
2	2	103	16.50
3	3	138	21.64
4	4	141	27.31
5	5	151	30.78
6	6	138	38.38
7	7	106	43.70
8	8	82	50.40
9	9	72	54.26
10	10	62	63.15
11	11	28	56.43
12	12	19	91.68
13	13	19	164.05
14	14	14	95.50
15	15	11	149.64
16	16	7	90.71
17	17	9	342.00

Interpretation: The results indicate an association between a greater number of distinct documented health conditions and higher healthcare utilization within the synthetic population. The relationship was not perfectly consistent across every condition-count group, particularly among groups containing relatively few patients. These findings describe an association and do not establish that additional health conditions caused greater healthcare utilization.

4.5 Conditions and Healthcare Expenses

Research Question: How is the number of distinct documented health conditions associated with healthcare expenses?

Method: The number of distinct documented conditions was calculated for each patient and joined with patient healthcare expense data using the unique patient identifier. Patients were grouped according to their number of unique conditions, and average healthcare expenses were calculated for each group. The analysis included 112 patients with matching records in both datasets.

	patients_in_both_tables
▶	112

Findings: Healthcare expenses generally tended to be higher among patients with a greater number of distinct documented conditions, although the pattern was not consistent across every condition-count group. For example, patients with one unique condition averaged \$46,005.06 in healthcare expenses, compared with \$610,352.55 among patients with five conditions and \$1,254,931.68 among patients with ten conditions. Higher condition-count groups showed greater variation and contained fewer patients.

	unique_conditions	patient_count	avg_healthcare_expenses
▶	1	2	460050.6
	2	13	417676.94
	3	18	591406.75
	4	13	836978.21
	5	15	610352.55
	6	19	834189.36
	7	7	925763.25
	8	6	804961.03
	9	7	754903.73
	10	5	1254931.68
	12	3	687994.08
	13	2	1386500.44
	14	1	1125499.22
	19	1	1075827.73

Interpretation: The results suggest an association between a greater number of distinct documented health conditions and higher healthcare expenses within the matched synthetic patient sample. However, the relationship was not strictly linear and estimates for groups with many conditions should be interpreted cautiously because some contained very few patients. These results demonstrate an association and do not establish that having additional conditions caused higher healthcare expenses.

KEY FINDINGS & CONCLUSION

KEY FINDINGS

The analysis identified several patterns across patient demographics, healthcare utilization, clinical conditions, and healthcare expenses.

Patients had an average age of 42.65 years, with representation across all five age groups. Healthcare expenses generally increased across older age groups, with patients aged 65 and older having the highest average expenses at approximately \$1.28 million.

A total of 53,346 healthcare encounters were analyzed. Wellness and ambulatory encounters were the most frequently recorded encounter types, while emergency and inpatient encounters occurred less frequently. Healthcare utilization also varied by age, with patients aged 65 and older averaging 49.75 encounters, compared with 16.32 encounters among patients aged 0–17.

Clinical analysis showed that viral sinusitis was the most frequently documented condition, followed by acute viral pharyngitis and acute bronchitis. Most patients represented in the conditions dataset had multiple distinct documented health conditions, with 1,115 of 1,152 patients (approximately 96.8%) having more than one distinct condition.

The cross-table analyses identified an overall association between greater numbers of distinct health conditions and increased healthcare utilization. Patients with one unique condition averaged 11.27 encounters, compared with 63.15 encounters among patients with ten

conditions. A similar overall pattern was observed for healthcare expenses: among the matched patient sample, patients with one unique condition averaged approximately \$46,005 in healthcare expenses, compared with approximately \$1.25 million among patients with ten conditions. These relationships were not perfectly linear and should be interpreted as associations rather than evidence of causation.

CONCLUSION

This analysis examined relationships among patient demographics, healthcare utilization, documented health conditions, and healthcare expenses using synthetic electronic health record data generated by Synthea. The findings identified differences in healthcare utilization and expenses across patient groups and showed that patients with greater numbers of distinct documented health conditions generally experienced higher healthcare utilization and healthcare expenses. However, these relationships were not consistent across every group and should be interpreted as associations rather than causal relationships.

Overall, the project demonstrates how SQL can be used to organize, query, and analyze healthcare data across multiple related tables. Techniques including aggregation, grouping, subqueries, distinct counts, and table joins were used to move from descriptive patient-level analysis to examining relationships across clinical, utilization, and expense data. Because the dataset is synthetic, the findings are intended to demonstrate analytical methods and patterns within the generated population rather than represent outcomes in a real-world patient population.